# Ericsson PE Router: BGP Route Churn, OSPF Adjacency Loss, and ISIS LSP Flooding Resolution Procedure

**Node:** YEL-PE-11 | **Vendor:** Ericsson | **Severity:** Warning | **Alarm Code:** ALARM_7245

## Description

The Routing Protocol Flap (BGP route churn, OSPF adjacency loss, ISIS LSP flooding) failure domain on the Ericsson Provider Edge (PE) router at Yelahanka Sub-station is a complex issue that arises due to the interplay of various protocol mechanics and hardware components. The BGP route churn is often caused by instability in the network, leading to repeated withdrawals and announcements of prefixes, thereby causing the router to continuously update its routing table. Similarly, the OSPF adjacency loss can occur due to issues such as interface flapping, authentication failures, or mismatched hello timers, resulting in the loss of neighbor relationships and subsequent routing instability. The ISIS LSP flooding, on the other hand, can be triggered by events like topology changes, interface failures, or router reloads, causing a surge in link-state packet (LSP) advertisements and potentially leading to network congestion. The Ericsson PE router, being a critical component in the network, requires prompt attention to these issues to prevent service disruptions and ensure seamless traffic flow. This procedure outlines the steps to diagnose and resolve these issues, leveraging the Ericsson CLI dialect to execute diagnostic and resolution commands.

## Pre-requisite Checks

- Verify the physical integrity of the router and its connections at Yelahanka Sub-station
- Check the status of redundant links to ensure failover capabilities are intact
- Confirm the availability of remote hands or dispatch personnel to the site if necessary

## Potential Root Causes

| Cause | Probability |
|---|---|
| BGP Route Churn due to Network Instability | High |
| OSPF Adjacency Loss due to Interface Issues | Medium |
| ISIS LSP Flooding due to Topology Changes | Low |

## Diagnostic Commands & Expected Outputs

```
Command: show ip bgp neighbor, Expected Output: BGP neighbor information, including state and last error, Description: This command isolates issues in the BGP control plane by displaying the state of BGP neighbors and any errors encountered, helping to identify potential causes of route churn.
```

```
Command: show ip ospf interface, Expected Output: OSPF interface information, including hello timers and neighbor count, Description: This command examines the OSPF interface configuration and operational state, aiding in the diagnosis of adjacency loss by highlighting potential mismatches in hello timers or other interface-related issues.

Command: show isis database, Expected Output: ISIS link-state database, including LSP information, Description: This command provides insight into the ISIS link-state database, allowing for the identification of potential flooding issues by examining LSP advertisements and their sources.
```

## Resolution Steps

Step 1: Step: BGP Route Churn Resolution, Command: neighbor route-map out, Description: Apply a route map to the BGP neighbor to filter out unstable prefixes, thereby reducing route churn. The route map should be configured to match the specific prefixes causing the instability.

Step 2: Step: OSPF Adjacency Loss Resolution, Command: interface ip ospf hello-interval , Description: Adjust the OSPF hello interval on the affected interface to match the neighboring router's configuration, ensuring consistent hello timers and preventing adjacency loss due to timer mismatches.

Step 3: Step: ISIS LSP Flooding Resolution, Command: isis network point-to-point, Description: Configure the ISIS network type to point-to-point on the affected interfaces to reduce LSP flooding by limiting the scope of LSP advertisements to directly connected neighbors.

## Post-Check Verification

```
Command: show ip bgp, Expected Output: BGP routing table, including prefixes and next hops, Description: Verify that the BGP routing table is stable and prefixes are being correctly advertised and received.

Command: show ip ospf neighbor, Expected Output: OSPF neighbor information, including state and dead timers, Description: Confirm that OSPF neighbors are in a full state and dead timers are appropriately set, indicating stable adjacencies.

Command: show isis spf-log, Expected Output: ISIS SPF calculation log, including timestamps and events, Description: Examine the ISIS SPF log to ensure that there are no recent SPF calculations due to LSP flooding, indicating a stable network topology.
```

## Roll-back Procedures

- Step: Revert BGP Route Map, Command: no neighbor route-map out, Description: Remove the applied route map from the BGP neighbor configuration to revert to the original state, should the fix introduce unforeseen issues.

- Step: Reset OSPF Hello Interval, Command: no interface ip ospf hello-interval, Description: Revert the OSPF hello interval to its default or previous value if the adjusted interval causes instability or other unforeseen effects.

- Step: Revert ISIS Network Type, Command: no isis network point-to-point, Description: Change the ISIS network type back to its original configuration if the point-to-point setting introduces issues with LSP flooding or neighbor discovery.